The effect of Test-Time Scaling for Visual Thinking Models

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IFT 6765
Links between Computer Vision and Language

MODELS: SAME 3B BACKBONE, DIFFERENT REASONING

Qwen2.5-VL-3B
Direct answer. No intermediate reasoning.
image + Q → ans

GRIT-3B
Grounded visual chain-of-thought.
image + Q → think → bbox → rethink → ans

HOW GRIT REASONS: WORKED EXAMPLE

Q: "What is the position of the cat in relation to the vase?" (A) behind (B) on top (C) inside (D) in front

input image

1
<think>
The cat appears to be resting inside or on top of the vase opening. I need to look more closely at the region around the cat and vase to disambiguate "on top" from "inside".
</think>

COMPUTE & EMIT BBOX `[108, 5, 236, 178]` → crop & re-examine

grounded crop

2
<rethink>
Looking at the crop, the cat is inside the lid opening, not in front or behind.
answer: B ✓

BENCHMARK: HARD, RECENT, VISION-INDISPENSABLE

MMMU-Pro · VQA | OCRBench v2 · OCR | MMStar · Counting

100 questions per task. All three datasets post-date model training cutoffs and require genuine visual grounding - text alone cannot answer them.

TTS RECIPE: 9 CANDIDATES PER QUESTION

Two diversity axes: temperature + image augmentation

- Original · T = 0 (greedy)
- Paraphrase A · T = 0.7
- Brightness · T = 0.7
- Edge enhance · T = 0.7
- **Rotation 90°** · T = 0.7
- Original · T = 0.7
- Paraphrase B · T = 0.7
- Grayscale · T = 0.7
- JPEG recompress · T = 0.7

Metric: **pass@9**. Correct if *any* of the 9 candidates is correct. Upper bound on what a perfect selector could recover.

HEADLINE: TTS LIFTS ACCURACY SHARPLY

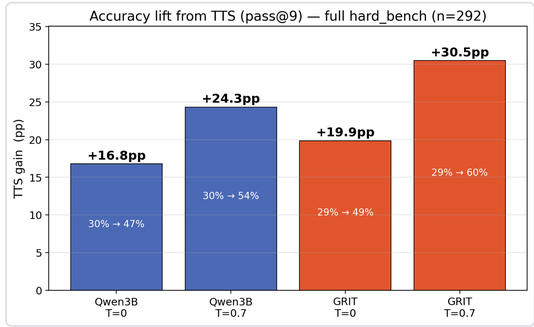

Fig. A - TTS gain (pass@9 – greedy) across all 4 configurations on the full 300-question benchmark. Standard recipe (T = 0.7 + augmentation) reaches **+30.5 pp for GRIT** and **+24.3 pp for Qwen3B**.

FINDING 1
Visual CoT benefits from TTS via image-input diversity; direct models don't.

At T = 0, Qwen produces the *same* answer regardless of augmentation. GRIT still gains **+19.9 pp** from those same augmentations: its grounding step amplifies small visual perturbations into different reasoning chains.

PER-TASK BREAKDOWN: GAINS LARGEST WHERE MODELS HAVE HEADROOM

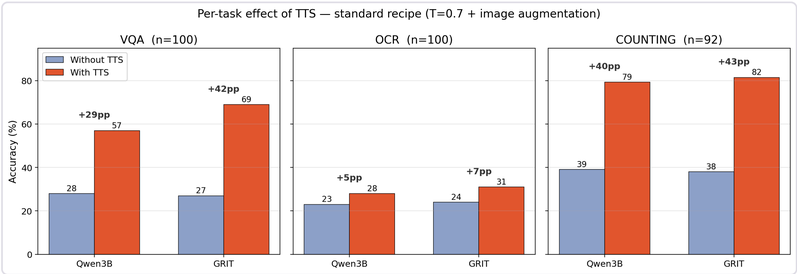

Fig. B - Per-task effect of TTS under the standard recipe. **VQA and Counting**, the two tasks where models have real headroom, show the largest gains (**+29 to +43 pp**). OCR gains stay modest (+5–7 pp) because exact-match scoring penalises augmentations that produce different-but-still-correct surface forms.

TAKE-HOME

Our results suggest TTS could benefit visual CoT models more than traditional VLMs. Visual grounding and chain-of-thought are powerful tools for generating genuinely diverse reasoning paths from input perturbations, paths a direct model cannot produce. **The open question is closing the gap between oracle and any practical candidate selection strategy.**

GRIT'S AUGMENTED CANDIDATES ARE COMPLEMENTARY, NOT REDUNDANT

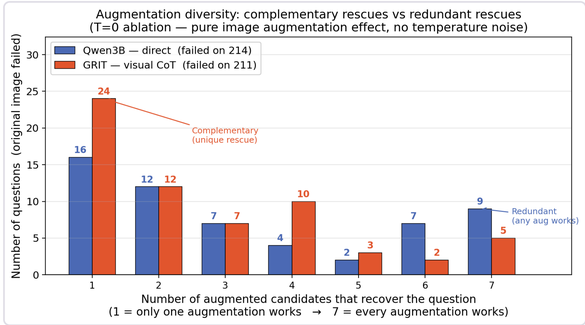

Fig. C - T = 0 ablation (no temperature noise). **GRIT** peaks at 1 unique rescue (complementary); **Qwen3B** clusters at 7 (redundant). Grounding turns spatial perturbations into structurally distinct chains.

VOTING CANNOT CLOSE THE ORACLE GAP

Plurality vote	Logprob sum	Greedy tiebreak	Consistency filter
−1.0 pp	**−3.1 pp**	**−1.7 pp**	**−1.0 pp**

FINDING 2
The diversity that raises the ceiling also makes aggregation hard.

GRIT's complementary rescues push oracle@9 high - each augmentation unlocks failures the others cannot. But that same spread defeats majority voting: no single candidate dominates. **The pool holds the signal; a learned verifier is needed to extract it.**

WHICH AUGMENTATION DRIVES THE GAIN?

Augmentation (T = 0)	GRIT uniq.	Qwen uniq.
Rotation 90°	10	4
JPEG recompress	5	3
Edge × Paraphrase	4	1
Brightness / contrast	3	4
Edge enhance	1	0
Paraphrase (text)	1	1
Grayscale	0	3

Unique rescue = questions where this augmentation is the **only** correct candidate. Rotation alters *spatial layout*; pixel-level augs do not.

★ GRIT's augmented candidates rescue **different** failures (complementary diversity); Qwen's rescue **the same** failures (redundant). Grounded reasoning turns spatial perturbations into structurally distinct chains; that is what makes TTS work.

LIMITATIONS
→ pass@9 is an oracle upper bound; no selector matches it yet
→ 3B models only; OCR uses strict exact-match scoring

NEXT STEPS
→ Confidence-based / learned verifier for selector design
→ DeepEyesV2 (agentic CoT) full evaluation

DATASETS · MODELS · COMPUTE
MMMU-Pro | OCRBench v2 | MMStar | Qwen2.5-VL-3B | GRIT-3B | Rorqual H100 · DRAC

SCAN FOR CODE